

Project Overview: Spatio-Temporal Variability of the Water Cycle

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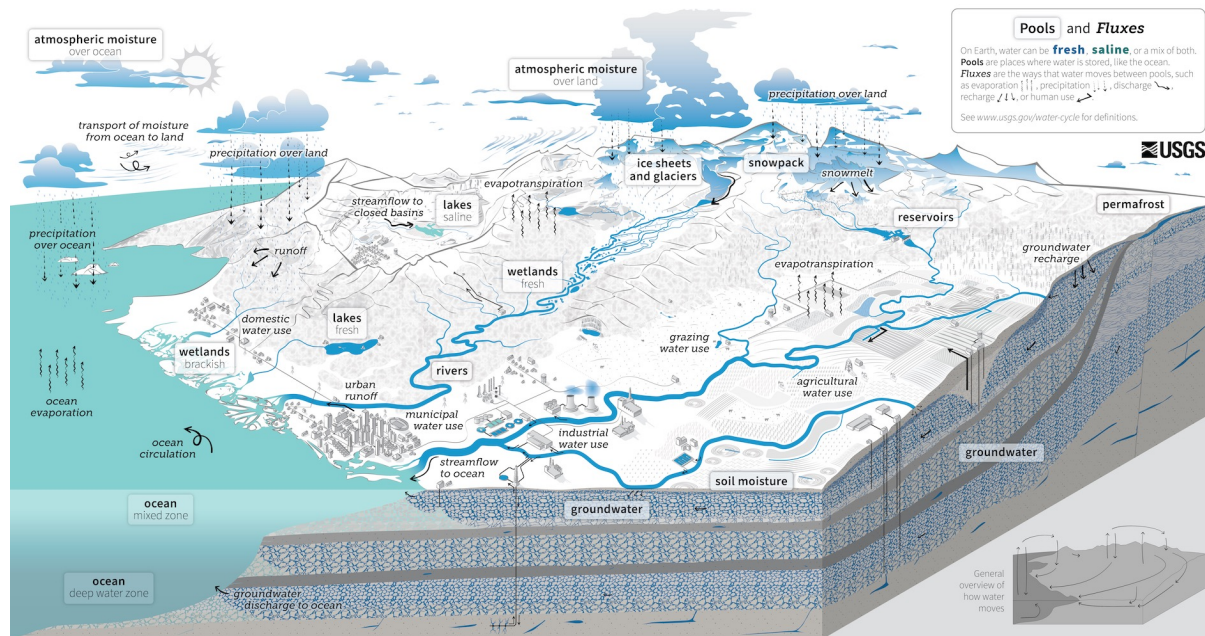
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Water Cycle



The Water Cycle

The water cycle describes where water is found on Earth and how it moves. Water can be stored in the atmosphere, on Earth's surface, or below the ground. It can be in a liquid, solid, or gaseous state. Water moves between the places it is stored at large scales and at very small scales. Water moves naturally and because of human interaction, both of which affect where water is stored, how it moves, and how clean it is.

Liquid water can be fresh, saline (salty), or a mix (brackish). Ninety-six percent of all water is saline and stored in **oceans**. Places like the ocean, where water is stored, are called **pools**. On land, saline water is stored in **saline lakes**, whereas fresh water is stored in liquid form in **freshwater lakes**, artificial **reservoirs**, **rivers**, **wetlands**, and in soil as **soil moisture**. Deeper underground, liquid water is stored as **groundwater** in aquifers, within the cracks and pores of rock. The solid, frozen form of water is stored in **ice sheets**, **glaciers**, and **snowpack** at high elevations or near the Earth's poles. Frozen water is also found in the soil as **permafrost**. Water vapor, the gaseous form of water, is stored as **atmospheric moisture** over the ocean and land.

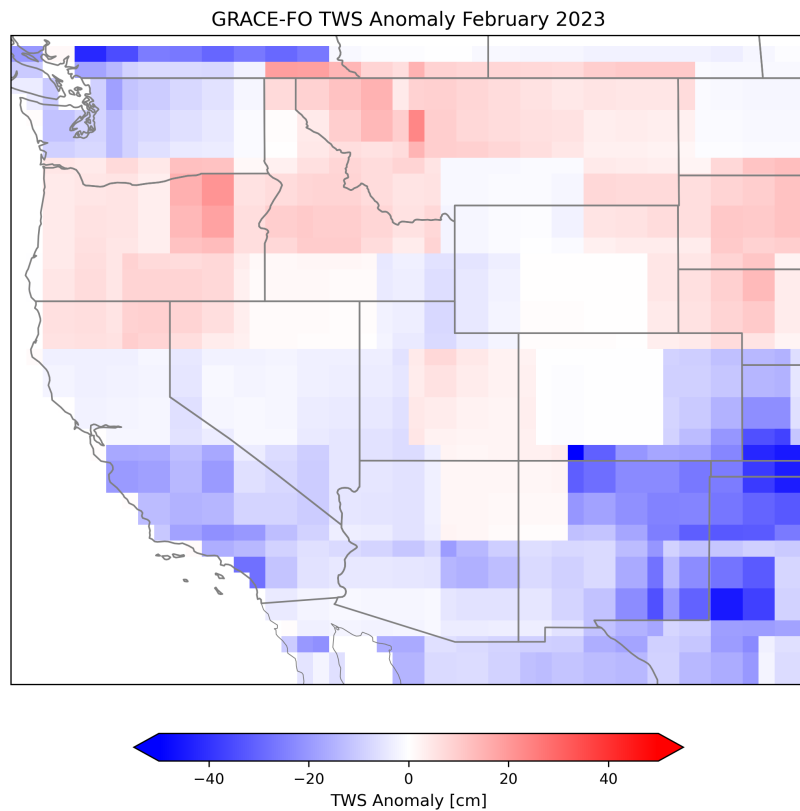
As it moves, water can transform into a liquid, a solid, or a gas. The different ways in which water moves between pools are known as **fluxes**. **Circulation** mixes water in the oceans and transports water vapor in the atmosphere. Water moves between the atmosphere and the Earth's surface through **evaporation**, **evapotranspiration**, and **precipitation**. Water moves across the land surface through **snowmelt**, **runoff**, and **streamflow**. Through infiltration and **groundwater recharge**, water moves into the ground. When underground, groundwater flows within aquifers and can return to the surface through **springs** or from natural **groundwater discharge** into rivers and oceans.

Humans alter the water cycle. We redirect rivers, build dams to store water, and drain water from wetlands for development. We use water from rivers, lakes, reservoirs, and groundwater aquifers. We use that water (1) to supply our **homes and communities**; (2) for **agricultural** irrigation and **grazing** livestock; and (3) in **industrial** activities like thermoelectric power generation, mining, and aquaculture. The amount of available water depends on how much water is in each pool (water quantity). Water availability also depends on when and how fast water moves (water timing), how much water is used (water use), and how clean the water is (water quality).

Human activities affect **water quality**. In agricultural and urban areas, irrigation and precipitation wash fertilizers and pesticides into rivers and groundwater. Power plants and factories return heated and contaminated water to rivers. Runoff carries chemicals, sediment, and sewage into rivers and lakes. Downstream from these types of sources, contaminated water can cause harmful algal blooms, spread diseases, and harm habitats. **Climate change** is also affecting the water cycle. It affects water quality, quantity, timing, and use. Climate change is also causing ocean acidification, sea level rise, and extreme weather. Understanding these impacts can allow progress toward sustainable water use.

- Continuous monitoring of the water cycle is critical for understanding climate system
- Multiple observing technologies, including remote sensing, employed to monitor different water cycle components
- Variability/changes in *storage pools* span a wide range of spatial and temporal scales

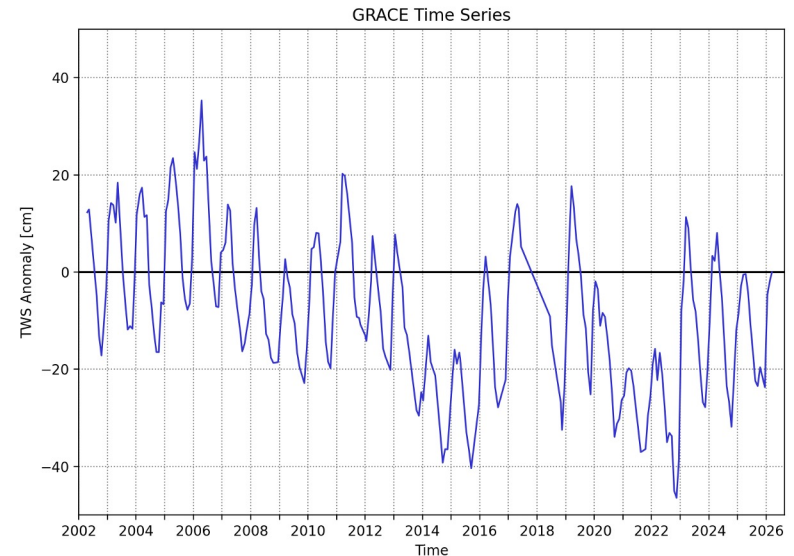
Project Learning Objectives



1. Gain familiarity with capabilities and limitations of NASA's multi-decade total water storage data record
2. Estimate and contrast long-term trends and seasonal variability in terrestrial water storage across regions/continents
3. Basic conceptual understanding of statistical models for spatial and temporal correlation

GRACE, GRACE-FO

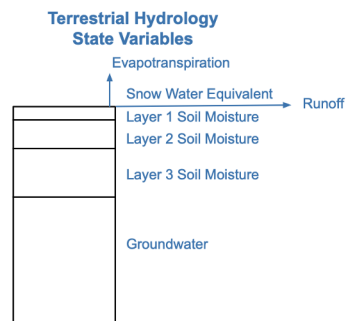
- The Gravity Recovery and Climate Experiment (GRACE) and follow-on (GRACE-FO) monitor Earth's gravity field
- Gravity field anomalies can be linked to multiple geophysical processes, including changes in terrestrial water storage: snow, soil moisture, groundwater
- Project will involve analysis of selected GRACE time series and spatial fields, applying statistical modeling at local, regional, and continental scales



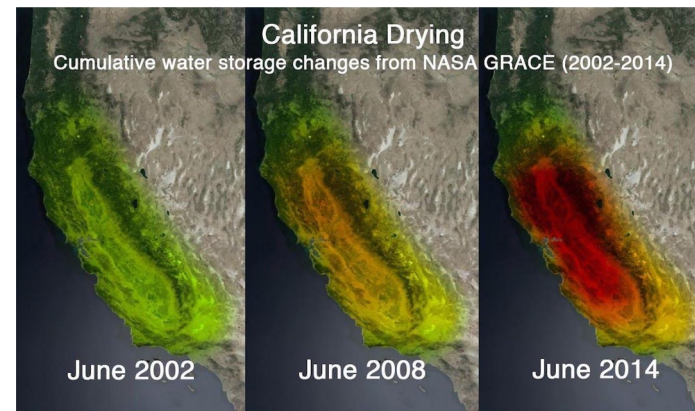
Monthly total water storage (TWS) anomalies estimated from the GRACE and GRACE-FO mission over central California
grace.jpl.nasa.gov

Terrestrial Hydrology Products

- Combined record of GRACE (2002-17) and GRACE-FO (2018-present) cover period of dramatic change in water use and availability
- Products give combined 2D spatial (~ 100 km) and temporal (~ monthly) snapshot of vertically integrated water storage
- Current operational products combine records of both missions:
<https://doi.org/10.5067/TEMSC-3JC634>



Typical representation of hydrology state variables in Earth system models. An estimate of the vertical aggregate is available from GRACE and GRACE-FO.



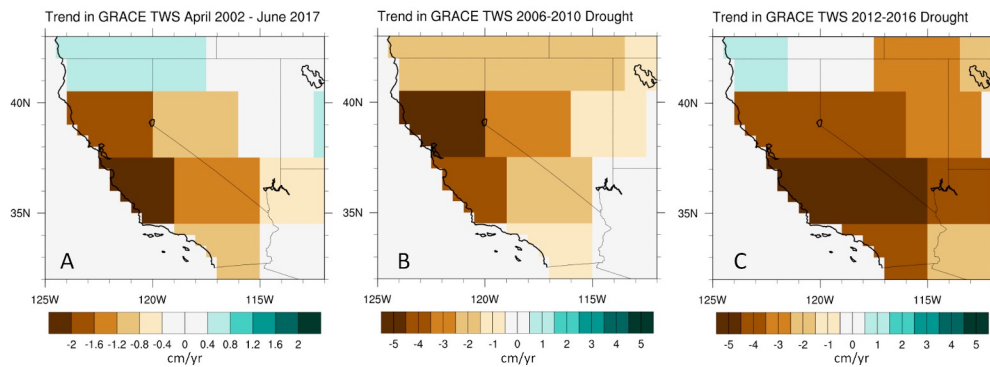
Snapshots of water storage over California estimated from GRACE
Source: <https://gracefo.jpl.nasa.gov/science/water-storage/>

Science Motivation

Exploration of data science capabilities motivated by multiple terrestrial hydrology objectives

1. Prediction of total water storage in data-limited periods (conditional inference)
2. Estimation and UQ for the annual cycle and long-term trends in water storage (unconditional inference)

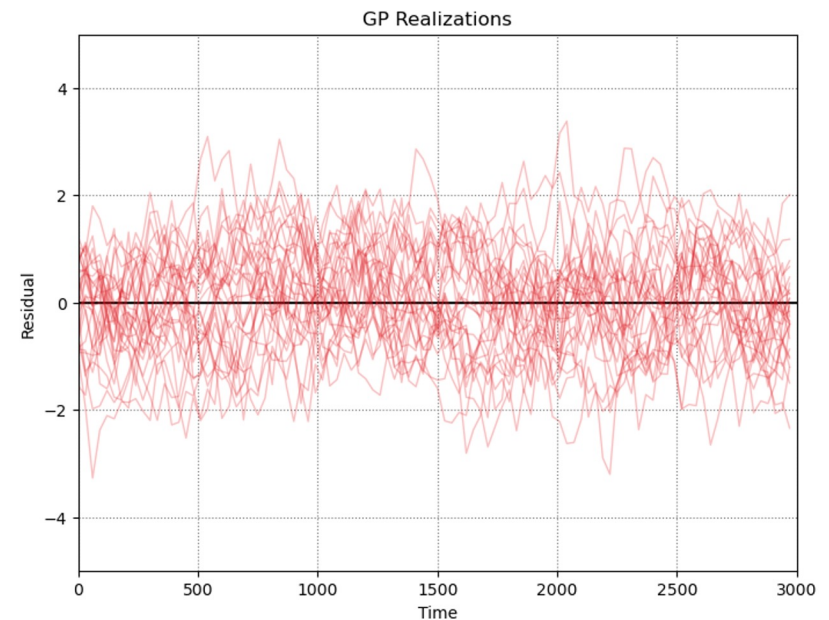
These objectives require accounting for and exploiting temporal correlation



E. Massoud, et al. (2022). "Cascading dynamics of the hydrologic cycle in California explored through observations and model simulations", *Geosciences*, 10. doi: 10.3390/geosciences10020071

Gaussian Processes

- Gaussian processes (GPs) are stochastic models, or probability distributions, for functions.
- Useful tool for representing uncertainty about a function that is incompletely observed by available data
 - Function has known input arguments, such as time, space
- Implementation of GPs goes by alternative names in different disciplines
 - *Kriging* in spatial statistics
 - *Gaussian stochastic processes* (GaSP) in UQ for computer models
 - *Gaussian process regression* (GPR) in machine learning



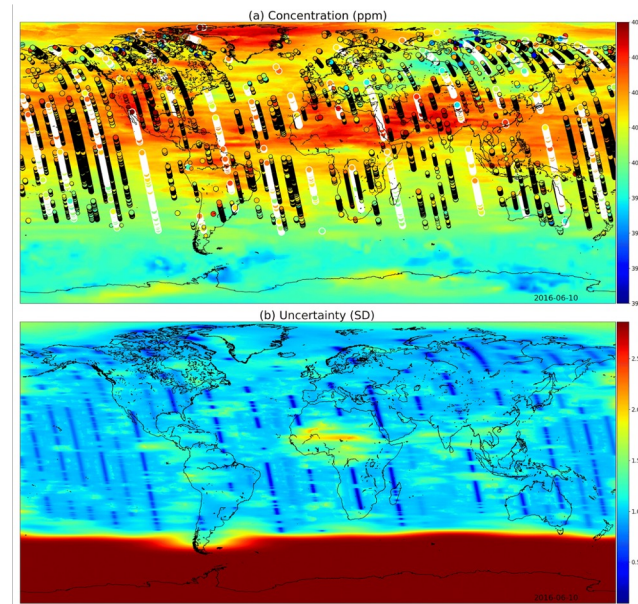
Gaussian Processes in Earth Science

Common Objective: Estimate properties of a “surface” or predict at unobserved locations given available data

Examples:

- Time series of terrestrial water storage (snow, soil moisture, groundwater) from multiple sensors
- Construction of spatially complete estimates of atmospheric composition
- Fast emulation of radiative transfer models for massive hyperspectral data

GPs and associated algorithms provide flexibility to address these objectives



Top: Estimated column CO₂ concentration from Susiluoto et al., 2020.

Bottom: Uncertainty estimate, all quantities have units of ppm

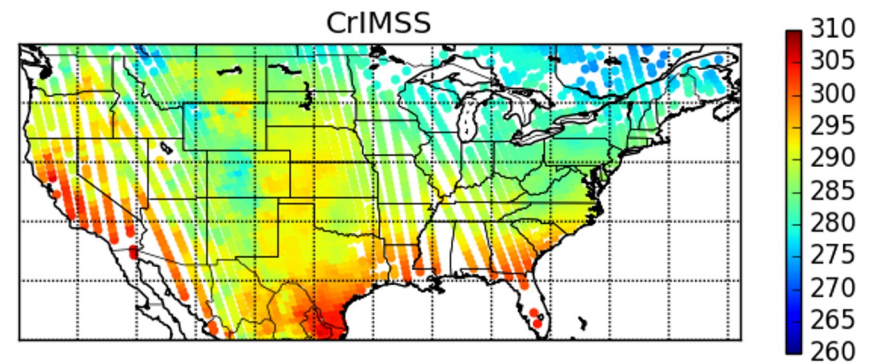
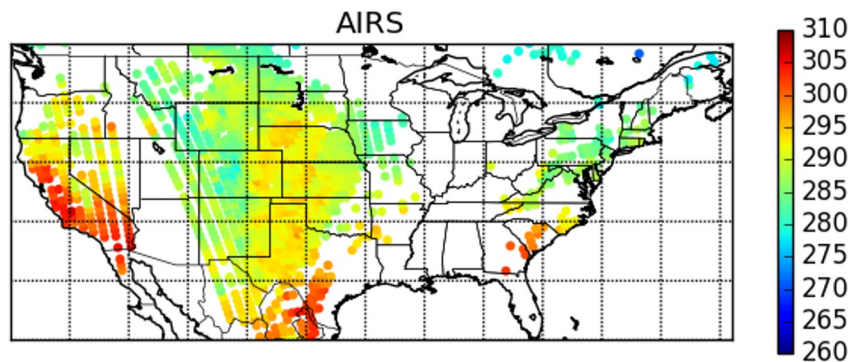
J. Susiluoto, et al. (2020). "Efficient multi-scale Gaussian process regression for massive remote sensing data with satGP v0.1.2", *Geosci. Model Dev.*, 13, 3439-3463. doi: 10.5194/gmd-13-3439-2020

Data Fusion

Goal: Regularly-spaced gap-filled estimates of quantity of interest from multiple remote sensing sources with complementary coverage

Challenges:

- Different statistical properties (bias, uncertainty) in input products
- Change of support/footprint size

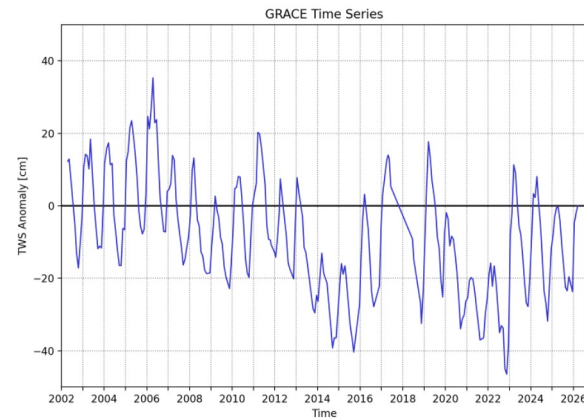
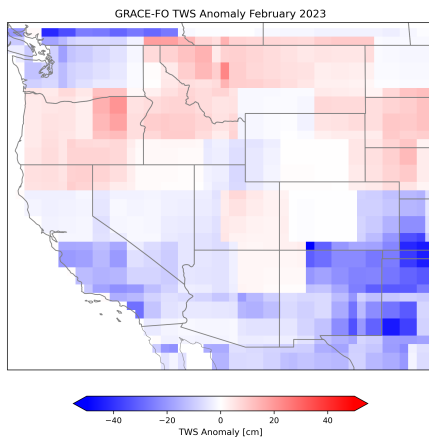


Left: Near-surface temperature retrievals from AIRS (in Kelvin) for a single day.
Right: Corresponding retrieved temperatures from CrIMSS

P. Kalmus, et al. (2022). "Data fusion of AIRS and CrIMSS near surface air temperature", Earth Space Sci., 9, e2022EA002282. doi: 10.1029/2022EA002282

Project Snapshot

- Example Python notebooks for GRACE/GRACE-FO
 - Extract full time series for location(s) of interest
 - Fit regression model for estimating trend and seasonal cycle
 - Fit Gaussian processes for residuals
 - Bootstrap for uncertainty quantification
- Extended analysis and questions
 - Regional analysis: Is a common seasonal cycle/trend sufficient for nearby locations within a region? How can spatial correlation of residuals be evaluated?
 - Contrast across regions/continents: Are temporal correlation length scales different across regions? What does that suggest about hydrologic processes in these areas?



Questions?

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Massoud, E., M. Turmon, J. Reager, J. Hobbs, Z. Liu, and C. David, 2020: Cascading dynamics of the hydrologic cycle in California explored through observations and model simulations. *Geosciences*, **10**, 71, <https://doi.org/10.3390/geosciences10020071>.

Susiluoto, J., A. Spantini, H. Haario, Härkönen, and Y. Marzouk, 2020: Efficient multi-scale Gaussian process regression for massive remote sensing data with satGP v0.1.2. *Geoscientific Model Development*, **13**, 3439–3463, <https://doi.org/10.5194/gmd-13-3439-2020>.